ALA-TOO INTERNATIONAL UNIVERSITY

BUSINESS ADMINISTRATION SOFTWARE

Spring Semester 2025-2026

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If you can't describe what you're doing as a process, you don't know what you're doing

– W. Edwards Deming

FROM SYSTEMS TO OPERATIONS

CONNECTION TO WEEK 1

Last week introduced enterprise systems. This week explains what those systems actually automate: structured business processes and repeatable workflows.

BUSINESS PROCESSES & DIGITAL WORKFLOWS

HOW WORK IS ORGANIZED AND EXECUTED

How organizations structure work through business processes and how software systems transform those processes into controlled digital workflows?

A BUSINESS PROCESS

DEFINITION

A business process is a structured sequence of activities designed to achieve a specific organizational goal, such as fulfilling an order, paying suppliers, or onboarding employees

CORE CHARACTERISTICS

HOW PROCESSES BEHAVE

Business processes are repeatable, goal-oriented, and measurable activities that define how work is performed within an organization. They involve multiple roles and departments that must coordinate to achieve a specific outcome, such as delivering a product, processing a payment, or supporting a customer

PROCESS EXECUTION AT SCALE

WHY SYSTEMS ARE REQUIRED

Because business processes are executed continuously and often at large scale, organizations standardize them and support them with digital systems. Software systems reduce errors, enforce consistency, maintain accountability, and enable organizations to monitor performance and improve processes over time

PROCESS IN PRACTICE

AMAZON

Order-to-Cash

Millions of daily orders processed through integrated systems covering inventory, warehousing, shipping, billing, and customer communication

AMAZON

ORDER FULFILLMENT PROCESS

Amazon’s order fulfillment process integrates customer ordering, inventory availability checks, warehouse picking and packing, shipping coordination, billing, and customer notifications. These activities are executed through tightly integrated software systems that ensure each step is triggered automatically and completed in the correct sequence.

AMAZON

PROCESS AT SCALE

Amazon processes millions of customer orders every day across global markets. This scale is achievable only because workflows are highly automated, standardized, and continuously monitored. Without digitally enforced processes, managing this volume with speed and accuracy would be impossible.

PROCESS IN PRACTICE

WALMART

Procure-to-Pay

Purchases and payments managed across 100,000+ suppliers using standardized ERP workflows.

WALMART

PROCURE-TO-PAY PROCESS

Walmart's procure-to-pay process manages how goods are purchased from suppliers, received into inventory, invoiced, and paid. This process connects procurement, inventory management, finance, and supplier systems, ensuring that purchasing activities follow standardized rules and approvals

WALMART

PROCESS AT SCALE

Walmart works with more than 100,000 suppliers worldwide, making manual purchasing and payment processes unmanageable. ERP-driven workflows allow Walmart to control spending, reduce errors, ensure timely payments, and maintain visibility across a highly complex global supply chain

PROCESS IN PRACTICE

IBM

Payroll & HR

Global workforce supported by structured HR and payroll processes ensuring accurate payments and regulatory compliance

IBM

PAYROLL & HR PROCESSES

IBM’s payroll and human resources processes manage employee records, compensation, benefits, and compliance across multiple countries. These processes rely on structured workflows to ensure that employee data is accurate, approvals are enforced, and payroll is processed consistently

IBM

PROCESS AT SCALE

IBM supports a global workforce of hundreds of thousands of employees, operating under different legal and regulatory environments. Digital HR and payroll workflows are essential to ensure accurate salary payments, regulatory compliance, and reliable reporting across regions

FUNCTIONAL VS CROSS-FUNCTIONAL

WHY SILOS FAIL

Most real business processes cross departmental boundaries rather than remaining within a single function. A single customer order, for example, typically involves sales for order creation, inventory for availability and fulfillment, finance for invoicing and payment, logistics for delivery, and reporting systems for monitoring performance and outcomes. This cross-functional nature makes coordination and system integration essential for reliable and efficient operations.

WHY PROCESSES BREAK

COMMON FAILURE POINTS

WHY PROCESSES BREAK

COMMON FAILURE POINTS

Business processes often fail due to unclear ownership, manual handoffs, duplicated data, inconsistent business rules, and poor system integration. When responsibilities are not clearly defined, tasks are delayed or missed, and when processes rely on manual steps or disconnected systems, errors and data inconsistencies increase. These weaknesses reduce efficiency, hide problems from management, and make processes difficult to control or improve.

DIGITAL WORKFLOWS

PROCESS EXECUTION BY SOFTWARE

A digital workflow is a business process that is executed, monitored, and controlled by software systems. It follows predefined steps, business rules, approvals, and system triggers that determine how tasks move from one stage to another, who is responsible at each step, and how data is processed. By automating execution and enforcing rules, digital workflows reduce manual errors, improve transparency, and ensure that processes are performed consistently across the organization.

MANUAL VS DIGITAL

KEY DIFFERENCE

Manual processes rely on emails, spreadsheets, and individual decisions, which makes them difficult to control, track, and standardize. In contrast, digital workflows rely on software systems that automatically log actions, enforce predefined business rules, manage approvals, and generate data in real time. This shift from manual to digital execution improves transparency, reduces errors, and enables organizations to monitor and optimize processes continuously

LIFE BEFORE ERP

HOW COMPANIES OPERATED

Before ERP systems, organizations managed their operations using isolated tools and manual coordination. Finance, inventory, sales, and HR worked independently, often relying on spreadsheets, emails, paper records, and department-specific software with no shared data foundation

DEPARTMENTAL SILOS

DISCONNECTED OPERATIONS

Each department maintained its own data, rules, and reports. Information moved slowly between teams through manual handoffs, creating delays, misunderstandings, and conflicting versions of the truth

STRUCTURAL LIMITATIONS

WHY THE OLD MODEL FAILED

As companies grew, this fragmented approach became unmanageable. Data duplication increased, reporting accuracy declined, and management lost real-time visibility into operations, making strategic decision-making unreliable.

OPERATIONAL RISKS

REAL CONSEQUENCES

Without integrated systems, companies faced inventory mismatches, delayed invoicing, payroll errors, compliance risks, and customer dissatisfaction. These were not technical failures, but structural process failures.

ENTERPRISE RESOURCE PLANNING

CORE DEFINITION

Enterprise Resource Planning (ERP) is an integrated information system that manages an organization’s core business processes through a shared database, standardized workflows, and centralized control mechanisms.

WHY ERP EXISTS

BUSINESS MOTIVATION

ERP systems exist to replace fragmented processes with unified workflows, eliminate data inconsistencies, enforce business rules, and provide leadership with a single, reliable view of organizational performance.

ERP AS A SYSTEM

NOT JUST SOFTWARE

ERP is NOT a single application, but a coordinated system of modules that represent how a business actually operates. It formalizes processes into software logic.

CORE ERP PRINCIPLE

SINGLE SOURCE OF TRUTH

All ERP modules share one central database. Data entered once is reused everywhere, ensuring consistency across finance, operations, HR, procurement, and reporting.

ERP IN ACTION

END-TO-END FLOW

When a sales order is created, ERP systems automatically update inventory, generate accounting entries, trigger procurement if needed, and make the transaction available for management reporting

ERP MODULE STRUCTURE

FUNCTIONAL DESIGN

Typical ERP modules include Finance, Procurement, Inventory, Sales, Manufacturing, and Human Resources. Modules are integrated by design, not connected as afterthoughts.

REAL ERP PROVIDERS

INDUSTRY STANDARDS

SAP S/4HANA (large enterprises)

Oracle ERP Cloud (enterprise & public sector)

Microsoft Dynamics 365 (mid-to-large companies)

Oracle NetSuite (cloud-first businesses)

Odoo (SMEs and modular adoption)

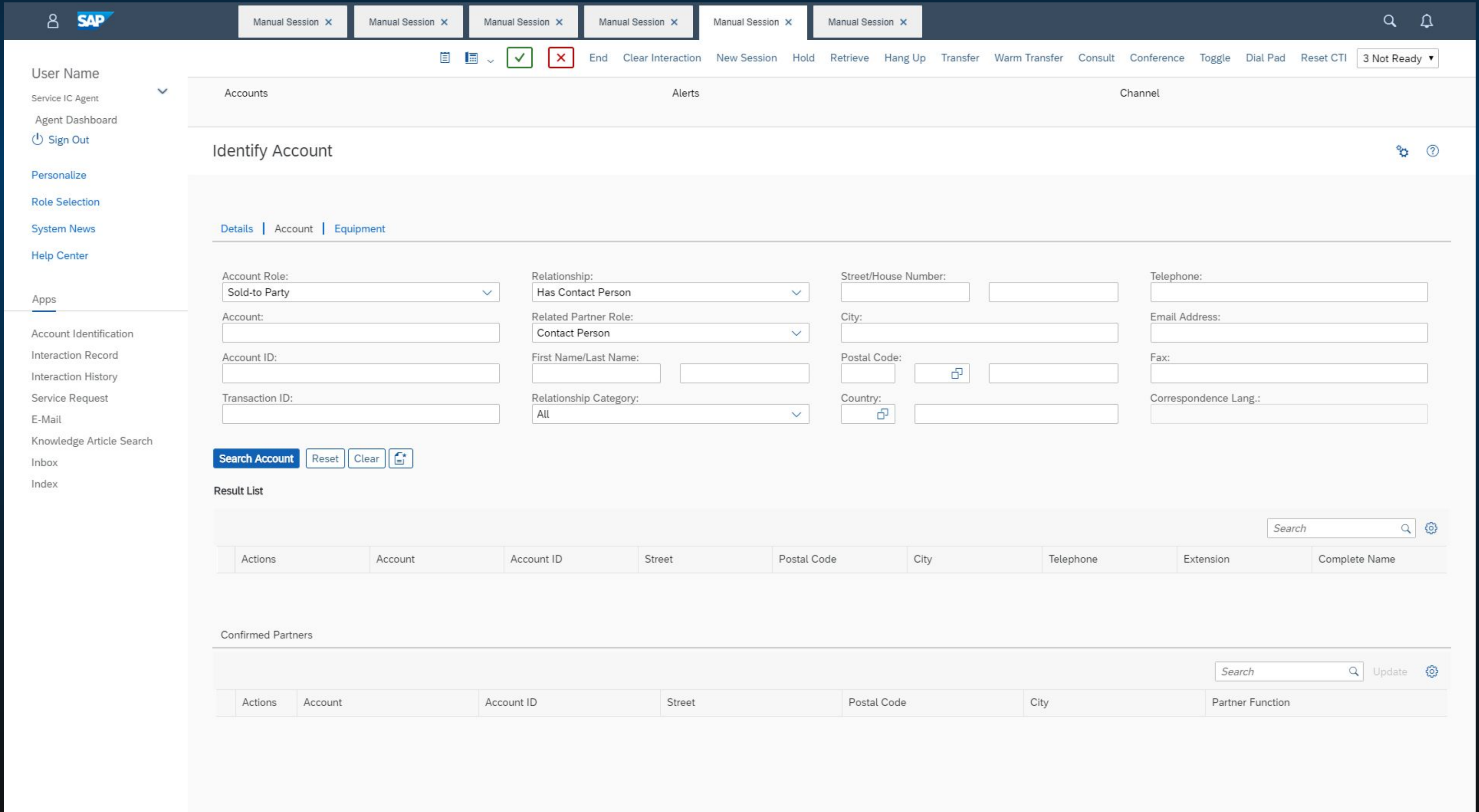
ERP AT SCALE

REAL WORLD USAGE

SAP systems process over 87% of global commerce by value, supporting millions of transactions daily across manufacturing, logistics, finance, and retail organizations.

SAP S/4HANA

REAL WORLD USAGE



ERP IMPLEMENTATION

MORE THAN INSTALLATION

ERP implementation involves process analysis, system configuration, data migration, integration, testing, and user training. Software alone does not fix broken processes.

MAINTENANCE & GOVERNANCE

ONGOING WORK

ERP systems require continuous governance: access control, performance monitoring, process optimization, compliance updates, and integration maintenance with external systems.

ERP AND IT SPECIALISTS

PROFESSIONAL ROLE

IT specialists configure ERP modules, manage integrations, ensure data integrity, enforce security rules, and translate business requirements into system behavior.

ERP FOR IT CAREERS

WHAT IT REALLY MEANS

ERP professionals must understand business logic, not just technology. Success depends on knowing how finance, operations, and data structures interact inside the system.

WHAT IF

ERP FAILS

Hershey's ERP rollout disrupted order processing and inventory coordination, leading to shipment delays and lost revenue during peak demand due to misaligned processes and insufficient testing.

FAILURE ANALYSIS

WHAT WENT WRONG

The system was technically functional, but business processes were poorly mapped, users were unprepared, and integrations were rushed. ERP failure was organizational, not technical.

ACTIVITY

ERP PROCESS TRACE

Trace a single business transaction through an ERP system to understand how one action impacts multiple modules and data records.

The goal is to build precise system-thinking skills rather than abstract discussion.

SCENARIO 0000

ORDER-TO-CASH FLOW

A customer places an order for 10 units of Product A at $100 per unit.
Current inventory level is 15 units.
The company uses an ERP system with integrated
sales, inventory and finance modules.

SCENARIO 0000

ORDER-TO-CASH FLOW

1. Which ERP modules are involved?
2. What happens to inventory quantity?
3. What accounting entries are created?
4. Whether procurement is triggered or not?

SCENARIO 0000

ORDER-TO-CASH FLOW

1. Modules involved: sales, inventory and finance

2. Inventory update: Stock decreases from 15 to 5 units

3. Accounting entry: Revenue of $1,000 recorded; accounts receivable created.

4. Procurement: Not triggered (sufficient inventory available)

STEP-BY-STEP VALIDATION

RECAP

Order: 10 units

Price: $100/unit

Inventory available: 15 units

STEP-BY-STEP VALIDATION

ERP MODULES INVOLVED

Sales → order creation

Inventory / Materials Management → stock check & reservation

Finance / Accounting → financial postings

STEP-BY-STEP VALIDATION

INVENTORY UPDATE

Two possible stages:

(1) At sales order creation → inventory is reserved

(2) At goods issue / shipment → inventory is physically reduced

Final result after shipment: Inventory goes from 15 → 5 units

That's standard ERP behavior

STEP-BY-STEP VALIDATION

ACCOUNTING ENTRY

At goods issue → Cost of Goods Sold (COGS) posted

At invoice creation → Revenue = $1,000

Accounts Receivable = $1,000

STEP-BY-STEP VALIDATION

PROCUREMENT TRIGGER

Inventory remaining after order: 5 units

Since stock ≥ 0 and no reorder rule mentioned → no procurement triggered

NEXT WEEK PREVIEW

WHAT'S COMING

Customer Relationship Management (CRM) systems